VIVEKANANDHA SAINIK COACHING CENTER

ALL INDIA SAINIK SCHOOL ENTRANCE EXAMINATION (AISSEE) CLASS IX

MOCK PAPER - 07

Time Allowed: 3 Hours (180 Mins) | **Maximum Marks:** 400 | **Total Questions:** 150

INSTRUCTIONS TO CANDIDATES:
1. This paper contains 150 questions divided into 5 Sections: Section A (Mathematics - 50 Qs, 200 Marks), Section B (Intelligence - 25 Qs, 50 Marks), Section C (English - 25 Qs, 50 Marks), Section D (General Science - 25 Qs, 50 Marks), Section E (Social Science - 25 Qs, 50 Marks).
2. All questions are compulsory. There is no negative marking.
3. Use standard black/blue ballpoint pen to darken OMR bubbles.

PART A: QUESTION PAPER

SECTION A: MATHEMATICS

Q1. What is the additive inverse of (-7/19)?

(a) 7/19 (b) -7/19
(c) 19/7 (d) -19/7

Q2. Solve for x: 3x - 5 = 2x + 7

(a) x = 2 (b) x = 12
(c) x = 6 (d) x = -12

Q3. The sum of three consecutive multiples of 8 is 888. Find the smallest multiple.

(a) 288 (b) 296
(c) 304 (d) 280

Q4. The measures of two adjacent angles of a parallelogram are in the ratio 3 : 2. Find the measure of the smaller angle.

(a) 108 degrees (b) 72 degrees
(c) 90 degrees (d) 60 degrees

Q5. How many diagonals does a regular hexagon have?

(a) 6 (b) 9
(c) 12 (d) 8

Q6. What is the minimum interior angle possible for a regular polygon?

(a) 30 degrees (b) 45 degrees
(c) 60 degrees (d) 90 degrees

Q7. In a pie chart representing the marks scored by a student, the central angle for Mathematics (80 marks out of 360 total marks) is:

(a) 80 degrees (b) 90 degrees
(c) 72 degrees (d) 100 degrees

Q8. A die is thrown once. What is the probability of getting a prime number?

(a) 1/2 (b) 1/3
(c) 1/6 (d) 2/3

Q9. Which of the following is a Pythagorean triplet whose smallest number is 8?

(a) (8, 15, 17) (b) (8, 12, 13)
(c) (8, 10, 12) (d) (8, 14, 16)

Q10. Find the square root of 729 by prime factorisation method.

(a) 23 (b) 27
(c) 29 (d) 31

Q11. Find the smallest natural number by which 256 must be multiplied so that the product becomes a perfect cube.

(a) 2 (b) 4
(c) 3 (d) 5

Q12. What is the value of cube root of 13824?

(a) 22 (b) 24
(c) 26 (d) 28

Q13. A shopkeeper bought a chair for Rs 1200 and sold it for Rs 1500. Find his profit percentage.

(a) 20% (b) 25%
(c) 30% (d) 15%

Q14. A jacket marked at Rs 2000 is sold for Rs 1600 after discount. What is the rate of discount?

(a) 15% (b) 20%
(c) 25% (d) 10%

Q15. Find the compound interest on Rs 10,000 for 2 years at 10% per annum compounded annually.

(a) Rs 2000 (b) Rs 2100
(c) Rs 1210 (d) Rs 2200

Q16. Subtract (4a - 7ab + 3b + 12) from (12a - 9ab + 5b - 3).

(a) 8a - 2ab + 2b - 15 (b) 8a + 2ab - 2b + 15
(c) 8a - 16ab + 8b + 9 (d) 16a - 2ab + 8b - 15

Q17. Using identity (a - b)^2 = a^2 - 2ab + b^2, find the value of (99)^2.

(a) 9801 (b) 9810
(c) 9701 (d) 9901

Q18. If Euler's formula for polyhedron is F + V - E = 2, find the number of edges (E) if Faces (F) = 6 and Vertices (V) = 8.

(a) 10 (b) 12
(c) 14 (d) 16

Q19. The area of a trapezium is 34 cm^2 and the length of one of the parallel sides is 10 cm and its height is 4 cm. Find the length of the other parallel side.

(a) 7 cm (b) 8 cm
(c) 6 cm (d) 9 cm

Q20. The diagonal of a rhombus are 7.5 cm and 12 cm. Find its area.

(a) 45 cm^2 (b) 90 cm^2
(c) 42 cm^2 (d) 50 cm^2

Q21. Find the total surface area of a closed cylinder of radius 7 cm and height 10 cm. (Use pi = 22/7)

(a) 748 cm^2 (b) 440 cm^2
(c) 308 cm^2 (d) 616 cm^2

Q22. Evaluate: (3^0 + 4^-1) x 2^2

(a) 5 (b) 4
(c) 3 (d) 2

Q23. Express 0.000035 in standard form.

(a) 3.5 x 10^-5 (b) 3.5 x 10^-4
(c) 35 x 10^-6 (d) 3.5 x 10^5

Q24. If 6 pipes fill a tank in 1 hour 20 minutes, how long will it take if only 5 pipes of the same type are used?

(a) 1 hour 36 minutes (b) 1 hour 30 minutes
(c) 1 hour 40 minutes (d) 2 hours

Q25. A machine in a soft drink factory fills 840 bottles in 6 hours. How many bottles will it fill in 5 hours?

(a) 700 (b) 720
(c) 650 (d) 750

Q26. Factorise fully: 15ab - 6a + 5b - 2

(a) (3a + 1)(5b - 2) (b) (3a - 1)(5b + 2)
(c) (5b - 2)(3a - 1) (d) (15a + 1)(b - 2)

Q27. Factorise: x^2 - 7x + 12

(a) (x - 3)(x - 4) (b) (x + 3)(x + 4)
(c) (x - 2)(x - 6) (d) (x + 1)(x - 12)

Q28. Divide: 44(x^4 - 5x^3 - 24x^2) by 11x(x - 8)

(a) 4x(x + 3) (b) 4x(x - 3)
(c) 4(x + 3) (d) 4x^2(x + 3)

Q29. If 21y5 is a multiple of 9, where y is a digit, what is the value of y?

(a) 1 (b) 2
(c) 3 (d) 0

Q30. The point (0, 5) lies on which axis?

(a) x-axis (b) y-axis
(c) Origin (d) Quadrant I

Q31. Simplify: (2^-1 x 4^-1) / 2^-2

(a) 1/2 (b) 1/4
(c) 1 (d) 2

Q32. Find the height of a cuboid whose base area is 180 cm^2 and volume is 900 cm^3.

(a) 5 cm (b) 6 cm
(c) 4 cm (d) 8 cm

Q33. A cuboid has dimensions 60 cm x 54 cm x 30 cm. How many small cubes with side 6 cm can be placed in the given cuboid?

(a) 450 (b) 500
(c) 400 (d) 480

Q34. By what number should (-15)^-1 be divided so that the quotient may be equal to (-5)^-1?

(a) 3 (b) 1/3
(c) -3 (d) -1/3

Q35. The ratio of ages of Anuj and Raj is 4 : 5. Eight years from now the ratio of their ages will be 5 : 6. Find Anuj's present age.

(a) 32 years (b) 40 years
(c) 24 years (d) 28 years

Q36. The sum of two numbers is 95. If one exceeds the other by 15, find the smaller number.

(a) 40 (b) 55
(c) 35 (d) 45

Q37. In a quadrilateral ABCD, angle A = 50 degrees, angle B = 60 degrees, angle C = 100 degrees. Find angle D.

(a) 150 degrees (b) 140 degrees
(c) 130 degrees (d) 120 degrees

Q38. How many sides does a regular polygon have if each of its interior angles is 165 degrees?

(a) 24 (b) 20
(c) 18 (d) 15

Q39. A group of 360 people were asked to vote for their favourite season from winter, rainy and summer. Summer got 90 votes, Rainy got 120 votes, Winter got 150 votes. Find central angle for Winter season.

(a) 150 degrees (b) 120 degrees
(c) 90 degrees (d) 100 degrees

Q40. What is the probability of drawing an Ace from a well-shuffled deck of 52 playing cards?

(a) 1/13 (b) 1/52
(c) 4/13 (d) 1/26

Q41. Find the square root of 12.25.

(a) 3.5 (b) 3.2
(c) 3.8 (d) 4.5

Q42. Is 53240 a perfect cube? If not, find the smallest number by which it must be divided to get a perfect cube.

(a) 5 (b) 2
(c) 4 (d) 10

Q43. Find the amount to be paid at the end of 1 year on Rs 8000 at 5% per annum compounded half-yearly.

(a) Rs 8405 (b) Rs 8400
(c) Rs 8410 (d) Rs 8500

Q44. Simplify the product: (a + b)(2a - 3b)

(a) 2a^2 - ab - 3b^2 (b) 2a^2 + ab - 3b^2
(c) 2a^2 - 3b^2 (d) 2a^2 + 3ab - 3b^2

Q45. The area of a rhombus is 240 cm^2 and one of the diagonals is 16 cm. Find the other diagonal.

(a) 30 cm (b) 20 cm
(c) 25 cm (d) 15 cm

Q46. Find the capacity of a cubical tank in litres if its inner edge is 2 m.

(a) 8000 L (b) 800 L
(c) 80000 L (d) 2000 L

Q47. Simplify: (3^-7 / 3^-10) x 3^-5

(a) 1/9 (b) 9
(c) 1/27 (d) 1/3

Q48. If 12 workers can build a wall in 40 hours, how many workers will be required to do the same work in 30 hours?

(a) 16 (b) 15
(c) 18 (d) 20

Q49. Factorise: 49x^2 - 36

(a) (7x - 6)(7x + 6) (b) (7x - 6)^2
(c) (7x + 6)^2 (d) (49x - 1)(x + 36)

Q50. If a 3-digit number 24x is divisible by 3, what are the possible values of x?

(a) 0, 3, 6, 9 (b) 1, 4, 7
(c) 2, 5, 8 (d) 3, 6, 9

SECTION B: INTELLIGENCE

Q51. If 'PUPIL' is coded as 'QVRJM', how is 'TEACHER' coded in that language?

(a) UFBDIFS (b) UFBDIES
(c) SFZBDDS (d) VFCEDFS

Q52. Find the missing number in the series: 2, 6, 12, 20, 30, ?

(a) 42 (b) 40
(c) 36 (d) 44

Q53. Select the odd one out:

(a) Square (b) Rectangle
(c) Rhombus (d) Sphere

Q54. Pointing to a photograph, a man said, 'I have no brother or sister but that man's father is my father's son.' Whose photograph was it?

(a) His son's (b) His father's
(c) His own (d) His nephew's

Q55. If '+' means multiplication, '-' means division, 'x' means subtraction and '/' means addition, then evaluate: 20 - 4 + 5 x 6 / 2

(a) 21 (b) 19
(c) 25 (d) 16

Q56. Complete the analogy: Doctor : Hospital :: Teacher : ?

(a) School (b) College
(c) Office (d) Student

Q57. A person walks 5 km North, turns Right and walks 3 km, then turns Right and walks 5 km. In which direction is he from his starting point?

(a) East (b) West
(c) North (d) South

Q58. Arrange the words in a logical order: 1. Country 2. Village 3. State 4. District 5. Continent

(a) 2, 4, 3, 1, 5 (b) 1, 2, 3, 4, 5
(c) 2, 3, 4, 1, 5 (d) 5, 1, 3, 4, 2

Q59. In a class of 40 students, Rohan is ranked 15th from the top. What is his rank from the bottom?

(a) 26th (b) 25th
(c) 27th (d) 24th

Q60. Find the missing letter: A, C, F, J, O, ?

(a) U (b) T
(c) V (d) W

Q61. Select the word which CANNOT be formed using the letters of the word 'DICTIONARY':

(a) NATION (b) RATION
(c) ACTION (d) TRAIN

Q62. Complete the analogy: 7 : 49 :: 9 : ?

(a) 81 (b) 72
(c) 90 (d) 63

Q63. If A = 1, CAT = 24, then DOG = ?

(a) 26 (b) 24
(c) 25 (d) 28

Q64. Which diagram best represents the relationship between: Animals, Dogs, Cats?

(a) Two disjoint circles inside a larger circle (b) Three intersecting circles
(c) Three concentric circles (d) Two intersecting circles inside a circle

Q65. Find the next term in the series: Z1A, X2D, V6G, T24J, ?

(a) R120M (b) S120M
(c) R100L (d) R120L

Q66. A cube is painted red on all faces and then cut into 64 smaller equal cubes. How many smaller cubes have exactly 3 faces painted?

(a) 8 (b) 12
(c) 24 (d) 16

Q67. Find the odd number pair out:

(a) 12 - 144 (b) 13 - 169
(c) 15 - 225 (d) 14 - 198

Q68. If BLUE is coded as 212215, how is RED coded?

(a) 1854 (b) 1845
(c) 1754 (d) 1954

Q69. If today is Monday, what day of the week will it be after 61 days?

(a) Saturday (b) Sunday
(c) Friday (d) Tuesday

Q70. At what angle are the hands of a clock inclined at 3:00?

(a) 90 degrees (b) 60 degrees
(c) 120 degrees (d) 180 degrees

Q71. Complete the non-verbal pattern series: Circle, Semi-circle, Quarter-circle, ?

(a) Point / Arc (b) Square
(c) Triangle (d) Oval

Q72. Select the correct mirror image of 'SAINIK' when mirror is placed to the right:

(a) Reversed letters right to left (b) Flipped vertically
(c) Same word (d) Inverted word

Q73. How many triangles are there in a pentagram (5-pointed star)?

(a) 10 (b) 8
(c) 5 (d) 12

Q74. Find the missing number in the grid: [[2, 3, 13], [4, 5, 41], [3, 6, ?]]

(a) 45 (b) 40
(c) 50 (d) 36

Q75. Choose the word which is least like the other words in the group:

(a) Copper (b) Zinc
(c) Iron (d) Brass

SECTION C: ENGLISH LANGUAGE

Q76. Read the passage and answer the question: 'Trees are vital for life on Earth. They absorb carbon dioxide and release oxygen, which is essential for all living beings. Forests also prevent soil erosion and maintain ecological balance.' — What do trees absorb according to the passage?

(a) Carbon dioxide (b) Oxygen
(c) Nitrogen (d) Hydrogen

Q77. What crucial function do forests perform regarding soil?

(a) Prevent soil erosion (b) Increase soil erosion
(c) Dry up soil (d) Remove minerals

Q78. What essential gas do trees release for living beings?

(a) Oxygen (b) Carbon dioxide
(c) Methane (d) Ozone

Q79. Which title is most appropriate for the passage?

(a) Importance of Trees (b) Types of Soil
(c) Gases in Atmosphere (d) Forest Fires

Q80. Identify the correctly spelt word:

(a) Accommodation (b) Acommodation
(c) Accomodation (d) Acomodation

Q81. Choose the correct synonym of 'BENEVOLENT':

(a) Kind (b) Cruel
(c) Greedy (d) Hateful

Q82. Choose the correct antonym of 'OPTISTIC':

(a) Pessimistic (b) Hopeful
(c) Positive (d) Confident

Q83. Fill in the blank with appropriate preposition: She has been living in Delhi _____ 2015.

(a) since (b) for
(c) from (d) in

Q84. Change into Passive Voice: 'The mechanic repaired the car.'

(a) The car was repaired by the mechanic. (b) The car is repaired by the mechanic.
(c) The car had repaired by mechanic. (d) The car repaired by mechanic.

Q85. Change into Indirect Speech: He said, 'I am reading a book.'

(a) He said that he was reading a book. (b) He said that I am reading a book.
(c) He said that he is reading a book. (d) He told he reads a book.

Q86. Fill in the blank with appropriate modal verb: You _____ follow the traffic rules strictly.

(a) must (b) might
(c) could (d) would

Q87. Identify the type of underlined clause: 'I know the place where he lives.'

(a) Adjective Clause (b) Noun Clause
(c) Adverb Clause (d) Prepositional Phrase

Q88. Fill in the blank: Neither he nor his friends _____ present at the party.

(a) were (b) was
(c) is (d) has

Q89. Complete the question tag: She sings very well, _____?

(a) doesn't she (b) does she
(c) isn't she (d) won't she

Q90. Choose the correct meaning of idiom: 'To spill the beans'

(a) To reveal a secret (b) To drop food
(c) To make a mistake (d) To work hard

Q91. Fill in the blank with correct article: Copper is _____ useful metal.

(a) a (b) an
(c) the (d) no article

Q92. Select the grammatically correct sentence:

(a) Each of the boys has submitted his project. (b) Each of the boys have submitted their project.
(c) Each of the boys are submitting project. (d) Each of boys have submitted.

Q93. Choose the word that means 'One who looks at the bright side of things':

(a) Optimist (b) Pessimist
(c) Atheist (d) Polyglot

Q94. Choose the antonym of 'TRANSPARENT':

(a) Opaque (b) Clear
(c) Lucid (d) Translucent

Q95. Fill in the blank: The sun _____ in the east.

(a) rises (b) is rising
(c) rose (d) has risen

Q96. Rearrange the jumbled words into a meaningful sentence: P: always / Q: honesty / R: best policy / S: is the

(a) Q-S-R-P / Honesty is always the best policy (b) P-Q-R-S
(c) S-R-Q-P (d) R-S-P-Q

Q97. Choose the correct conjunction: Work hard _____ you should fail.

(a) lest (b) unless
(c) otherwise (d) or

Q98. Identify the abstract noun in the sentence: 'His bravery earned him a medal.'

(a) Bravery (b) Medal
(c) His (d) Earned

Q99. Fill in the blank: The police _____ investigating the crime.

(a) are (b) is
(c) was (d) has

Q100. Choose the one-word substitution for: 'A list of books available in a library'

(a) Catalogue (b) Bibliography
(c) Index (d) Biography

SECTION D: GENERAL SCIENCE

Q101. Kharif crops are sown during which months in India?

(a) June to September (b) October to March
(c) November to April (d) January to May

Q102. Which microorganism is responsible for converting milk into curd?

(a) Lactobacillus (b) Rhizobium
(c) Yeast (d) Amoeba

Q103. Which synthetic fibre is known as 'artificial silk'?

(a) Rayon (b) Nylon
(c) Polyester (d) Acrylic

Q104. Which metal is liquid at room temperature?

(a) Mercury (b) Sodium
(c) Gallium (d) Bromine

Q105. Which gas is produced when metals react with acids?

(a) Hydrogen gas (b) Oxygen gas
(c) Carbon dioxide (d) Nitrogen gas

Q106. Which fossil fuel is known as 'black gold'?

(a) Petroleum (b) Coal
(c) Natural Gas (d) Coke

Q107. What is the ignition temperature of a fuel?

(a) Minimum temperature at which it catches fire (b) Maximum temperature it can reach
(c) Temperature at room level (d) Boiling point of fuel

Q108. The process of conversion of wood into coal under high pressure and temperature over long periods is called:

(a) Carbonisation (b) Destructive distillation
(c) Fractional distillation (d) Fossilisation

Q109. Which unit of biosphere conservation aims to preserve wildlife, flora, fauna, and traditional tribal life?

(a) Biosphere Reserve (b) National Park
(c) Wildlife Sanctuary (d) Zoo

Q110. The 'powerhouse of the cell' is:

(a) Mitochondria (b) Nucleus
(c) Ribosome (d) Golgi apparatus

Q111. Binary fission is a method of asexual reproduction observed in:

(a) Amoeba (b) Hydra
(c) Yeast (d) Human

Q112. Which hormone regulates blood sugar levels in the human body?

(a) Insulin (b) Thyroxin
(c) Adrenaline (d) Estrogen

Q113. The force exerted by a charged body on another charged or uncharged body is called:

(a) Electrostatic force (b) Gravitational force
(c) Magnetic force (d) Friction

Q114. Friction can be reduced by using substances called:

(a) Lubricants (b) Abrasives
(c) Adhesives (d) Insulators

Q115. The loudness of sound depends on its:

(a) Amplitude (b) Frequency
(c) Speed (d) Time period

Q116. The audible frequency range for human ears is approximately:

(a) 20 Hz to 20,000 Hz (b) Less than 20 Hz
(c) Above 20,000 Hz (d) 200 Hz to 2,000 Hz

Q117. Which of the following liquids is a good conductor of electricity?

(a) Lemon juice (b) Distilled water
(c) Vegetable oil (d) Sugar solution

Q118. The process of depositing a layer of any desired metal on another material by means of electricity is called:

(a) Electroplating (b) Galvanisation
(c) Electrolysis (d) Anodising

Q119. Like charges _____ each other, while unlike charges _____ each other.

(a) repel; attract (b) attract; repel
(c) neutralise; attract (d) repel; neutralise

Q120. The point on the Earth's surface directly above the seismic focus of an earthquake is called:

(a) Epicentre (b) Hypocentre
(c) Fault line (d) Seismic zone

Q121. An image formed by a plane mirror is always:

(a) Virtual, erect, and same size (b) Real, inverted, and enlarged
(c) Virtual, inverted, and diminished (d) Real, erect, and same size

Q122. The angle of incidence is equal to the angle of reflection according to:

(a) Laws of Reflection (b) Refraction Rule
(c) Dispersion Law (d) Snell's Law

Q123. Which planet is known as the 'Morning Star' or 'Evening Star'?

(a) Venus (b) Mars
(c) Jupiter (d) Mercury

Q124. Which celestial body is a natural satellite of the Earth?

(a) Moon (b) Sun
(c) Venus (d) Halley's Comet

Q125. Which gas is primarily responsible for global warming and greenhouse effect?

(a) Carbon dioxide (b) Oxygen
(c) Nitrogen (d) Argon

SECTION E: SOCIAL SCIENCE

Q126. Who was the first Governor-General of British India under the East India Company rule?

(a) Warren Hastings (b) Lord Dalhousie
(c) Lord Mountbatten (d) Robert Clive

Q127. The Battle of Plassey was fought in which year?

(a) 1757 (b) 1764
(c) 1857 (d) 1707

Q128. Under the Permanent Settlement introduced by Lord Cornwallis in 1793, who were recognized as zamindars?

(a) Rajas and Taluqdars (b) Peasants
(c) Village headmen (d) British officers

Q129. Who led the Revolt of 1857 in Kanpur?

(a) Nana Saheb (b) Rani Lakshmibai
(c) Begum Hazrat Mahal (d) Kunwar Singh

Q130. Who founded the Brahmo Samaj in 1828 to promote social reform?

(a) Raja Rammohan Roy (b) Swami Dayanand Saraswati
(c) Jyotirao Phule (d) Ishwar Chandra Vidyasagar

Q131. The Rowlatt Act was passed by the British Government in which year?

(a) 1919 (b) 1905
(c) 1930 (d) 1942

Q132. Which movement was launched by Mahatma Gandhi following the Dandi Salt March in 1930?

(a) Civil Disobedience Movement (b) Non-Cooperation Movement
(c) Quit India Movement (d) Swadeshi Movement

Q133. Resources that are found everywhere on Earth (like air) are classified as:

(a) Ubiquitous resources (b) Localized resources
(c) Potential resources (d) Non-renewable resources

Q134. Which layer of soil contains humus and fine organic matter essential for plant growth?

(a) Topsoil (b) Subsoil
(c) Weathered rock material (d) Bedrock

Q135. Terrace farming is an effective soil conservation method used primarily on:

(a) Steep mountain slopes (b) Flat river plains
(c) Desert sand dunes (d) Coastal beaches

Q136. Growing crops alongside rearing of livestock on the same farm is known as:

(a) Mixed farming (b) Subsistence farming
(c) Shifting cultivation (d) Plantation agriculture

Q137. Which region in India is famous for being a major steel and mineral industrial hub?

(a) Chota Nagpur Plateau (b) Deccan Plateau
(c) Malwa Plateau (d) Ladakh Plateau

Q138. The Supreme Court of India was established in Delhi in which year?

(a) 1950 (b) 1947
(c) 1858 (d) 1935

Q139. Which Article of the Indian Constitution abolishes 'Untouchability'?

(a) Article 17 (b) Article 14
(c) Article 19 (d) Article 21

Q140. The maximum sanctioned strength of members in the Lok Sabha is:

(a) 550 (b) 250
(c) 545 (d) 500

Q141. Who is the ex-officio Chairman of the Rajya Sabha in India?

(a) Vice-President of India (b) President of India
(c) Prime Minister (d) Speaker of Lok Sabha

Q142. The fundamental right that guarantees protection of life and personal liberty is enshrined in:

(a) Article 21 (b) Article 19
(c) Article 14 (d) Article 32

Q143. Public Interest Litigation (PIL) was introduced in India to:

(a) Increase access to justice for the poor (b) Speed up corporate trials
(c) Replace trial courts (d) Reduce tax disputes

Q144. Shifting cultivation in North-East India is locally known as:

(a) Jhumming (b) Milpa
(c) Roca (d) Ladang

Q145. Which industry is often termed as the 'backbone of modern industry'?

(a) Iron and Steel Industry (b) Textile Industry
(c) Information Technology (d) Petrochemical Industry

Q146. Who wrote the famous book 'Gulamgiri' in 1873 criticizing the caste system?

(a) Jyotirao Phule (b) BR Ambedkar
(c) Periyar EV Ramasamy (d) Sri Narayana Guru

Q147. Which Constitutional Amendment Act made elementary education a Fundamental Right under Article 21A?

(a) 86th Amendment Act (b) 44th Amendment Act
(c) 42nd Amendment Act (d) 73rd Amendment Act

Q148. Silicon Valley in California and Whitefield/Electronic City in Bengaluru are renowned centers for which industry?

(a) Information Technology (b) Cotton Textile
(c) Automobile (d) Shipbuilding

Q149. The doctrine of 'Lapse' was aggressively implemented by which British Governor-General?

(a) Lord Dalhousie (b) Lord Wellesley
(c) Lord Canning (d) Lord Curzon

Q150. What is the term duration for a member of the Rajya Sabha in India?

(a) 6 years (b) 5 years
(c) 4 years (d) Permanent without term

PART B: MASTER ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
Q1	a	Q2	b	Q3	a	Q4	b	Q5	b
Q6	c	Q7	a	Q8	a	Q9	a	Q10	b
Q11	a	Q12	b	Q13	b	Q14	b	Q15	b
Q16	a	Q17	a	Q18	b	Q19	a	Q20	a
Q21	a	Q22	a	Q23	a	Q24	a	Q25	a
Q26	a	Q27	a	Q28	a	Q29	a	Q30	b

Q31	a	Q32	a	Q33	a	Q34	a	Q35	a
Q36	a	Q37	a	Q38	a	Q39	a	Q40	a
Q41	a	Q42	a	Q43	a	Q44	a	Q45	a
Q46	a	Q47	a	Q48	a	Q49	a	Q50	a
Q51	a	Q52	a	Q53	d	Q54	a	Q55	a
Q56	a	Q57	a	Q58	a	Q59	a	Q60	a
Q61	a	Q62	a	Q63	a	Q64	a	Q65	a
Q66	a	Q67	d	Q68	a	Q69	a	Q70	a
Q71	a	Q72	a	Q73	a	Q74	a	Q75	d
Q76	a	Q77	a	Q78	a	Q79	a	Q80	a
Q81	a	Q82	a	Q83	a	Q84	a	Q85	a
Q86	a	Q87	a	Q88	a	Q89	a	Q90	a
Q91	a	Q92	a	Q93	a	Q94	a	Q95	a
Q96	a	Q97	a	Q98	a	Q99	a	Q100	a
Q101	a	Q102	a	Q103	a	Q104	a	Q105	a
Q106	a	Q107	a	Q108	a	Q109	a	Q110	a
Q111	a	Q112	a	Q113	a	Q114	a	Q115	a
Q116	a	Q117	a	Q118	a	Q119	a	Q120	a
Q121	a	Q122	a	Q123	a	Q124	a	Q125	a
Q126	a	Q127	a	Q128	a	Q129	a	Q130	a
Q131	a	Q132	a	Q133	a	Q134	a	Q135	a
Q136	a	Q137	a	Q138	a	Q139	a	Q140	a
Q141	a	Q142	a	Q143	a	Q144	a	Q145	a
Q146	a	Q147	a	Q148	a	Q149	a	Q150	a

PART C: DETAILED SOLUTIONS

SECTION A: MATHEMATICS — SOLUTIONS

Q1. Correct Option: (a)
The additive inverse of a number 'a' is '-a' such that a + (-a) = 0. Therefore, the additive inverse of (-7/19) is 7/19.

Q2. Correct Option: (b)
3x - 2x = 7 + 5 => x = 12.

Q3. Correct Option: (a)
Let the multiples be x, x+8, x+16. Their sum: 3x + 24 = 888 => 3x = 864 => x = 288. Smallest multiple is 288.

Q4. Correct Option: (b)
Adjacent angles of a parallelogram are supplementary. 3x + 2x = 180 => 5x = 180 => x = 36 degrees. Smaller angle = 2x = 72 degrees.

Q5. Correct Option: (b)
Number of diagonals in an n-sided polygon = n(n - 3) / 2. For hexagon (n = 6): 6(6 - 3)/2 = 6 x 3 / 2 = 9.

Q6. Correct Option: (c)
An equilateral triangle (n = 3) is the regular polygon with the fewest sides. Its interior angle is 60 degrees, which is the minimum.

Q7. Correct Option: (a)
Central angle = (Marks obtained / Total marks) x 360 degrees = (80 / 360) x 360 degrees = 80 degrees.

Q8. Correct Option: (a)
Outcomes = {1, 2, 3, 4, 5, 6}. Prime numbers = {2, 3, 5} (total 3). Probability = 3/6 = 1/2.

Q9. Correct Option: (a)
8^2 + 15^2 = 64 + 225 = 289 = 17^2. Thus (8, 15, 17) is a Pythagorean triplet.

Q10. Correct Option: (b)
729 = 3 x 3 x 3 x 3 x 3 x 3 = (3 x 3 x 3)^2 = 27^2. Hence, sqrt(729) = 27.

Q11. Correct Option: (a)
256 = 2^8 = (2^3) x (2^3) x (2^2). To make it a perfect cube, we need one more 2. Smallest multiplier = 2.

Q12. Correct Option: (b)
13824 = 2^9 x 3^3. Cube root = 2^3 x 3 = 8 x 3 = 24.

Q13. Correct Option: (b)
Profit = 1500 - 1200 = Rs 300. Profit % = (300 / 1200) x 100 = 25%.

Q14. Correct Option: (b)
Discount = 2000 - 1600 = Rs 400. Discount % = (400 / 2000) x 100 = 20%.

Q15. Correct Option: (b)
A = P(1 + R/100)^n = 10000(1 + 10/100)^2 = 10000 x (11/10)^2 = 12100. CI = 12100 - 10000 = Rs 2100.

Q16. Correct Option: (a)
(12a - 9ab + 5b - 3) - (4a - 7ab + 3b + 12) = 12a - 4a - 9ab + 7ab + 5b - 3b - 3 - 12 = 8a - 2ab + 2b - 15.

Q17. Correct Option: (a)
(99)^2 = (100 - 1)^2 = 10000 - 200 + 1 = 9801.

Q18. Correct Option: (b)
F + V - E = 2 => 6 + 8 - E = 2 => 14 - E = 2 => E = 12.

Q19. Correct Option: (a)
Area = (1/2) x (a + b) x h => 34 = (1/2) x (10 + b) x 4 => 34 = 2(10 + b) => 17 = 10 + b => b = 7 cm.

Q20. Correct Option: (a)
Area of rhombus = (1/2) x d1 x d2 = (1/2) x 7.5 x 12 = 45 cm^2.

Q21. Correct Option: (a)
TSA = 2 x pi x r x (r + h) = 2 x (22/7) x 7 x (7 + 10) = 44 x 17 = 748 cm^2.

Q22. Correct Option: (a)
(1 + 1/4) x 4 = (5/4) x 4 = 5.

Q23. Correct Option: (a)
0.000035 = 3.5 x 10^-5.

Q24. Correct Option: (a)
Number of pipes and time taken are inversely proportional. 6 x 80 mins = 5 x T => T = 480 / 5 = 96 mins = 1 hr 36 mins.

Q25. Correct Option: (a)
Direct proportion: 840 / 6 = x / 5 => 140 x 5 = 700 bottles.

Q26. Correct Option: (a)
15ab - 6a + 5b - 2 = 3a(5b - 2) + 1(5b - 2) = (3a + 1)(5b - 2).

Q27. Correct Option: (a)
x^2 - 7x + 12 = x^2 - 3x - 4x + 12 = (x - 3)(x - 4).

Q28. Correct Option: (a)
Numerator = 44 x x^2(x^2 - 5x - 24) = 44x^2(x - 8)(x + 3). Divide by 11x(x - 8) => 4x(x + 3).

Q29. Correct Option: (a)
Sum of digits = 2 + 1 + y + 5 = 8 + y. For multiple of 9, 8 + y = 9 => y = 1.

Q30. Correct Option: (b)
Since the x-coordinate is 0, the point lies on the y-axis.

Q31. Correct Option: (a)
(1/2 x 1/4) / (1/4) = (1/8) / (1/4) = (1/8) x 4 = 1/2.

Q32. Correct Option: (a)
Volume = Base Area x Height => 900 = 180 x h => h = 5 cm.

Q33. Correct Option: (a)
Number of cubes = (60 x 54 x 30) / (6 x 6 x 6) = 10 x 9 x 5 = 450.

Q34. Correct Option: (a)
(-1/15) / x = (-1/5) => x = (-1/15) / (-1/5) = (-1/15) x (-5) = 1/3? Wait, (-1/15) / x = -1/5 => x = (-1/15) * (-5) = 1/3.

Q35. Correct Option: (a)
Let ages be 4x and 5x. (4x + 8)/(5x + 8) = 5/6 => 24x + 48 = 25x + 40 => x = 8. Anuj's age = 4x = 32 years.

Q36. Correct Option: (a)
Let smaller number = x. Then x + (x + 15) = 95 => 2x = 80 => x = 40.

Q37. Correct Option: (a)
Sum of angles in a quadrilateral = 360 degrees. Angle D = 360 - (50 + 60 + 100) = 360 - 210 = 150 degrees.

Q38. Correct Option: (a)
Each exterior angle = 180 - 165 = 15 degrees. Number of sides n = 360 / 15 = 24.

Q39. Correct Option: (a)
Central angle = (150 / 360) x 360 degrees = 150 degrees.

Q40. Correct Option: (a)
Number of Aces = 4. Total cards = 52. Probability = 4 / 52 = 1 / 13.

Q41. Correct Option: (a)
3.5 x 3.5 = 12.25. So sqrt(12.25) = 3.5.

Q42. Correct Option: (a)
53240 = 2^3 x 11^3 x 5. Dividing by 5 gives 2^3 x 11^3, which is a perfect cube. Smallest divisor = 5.

Q43. Correct Option: (a)
Rate per half year = 2.5%, n = 2 periods. A = 8000 x (1 + 2.5/100)^2 = 8000 x (41/40)^2 = 8000 x 1681 / 1600 = 5 x 1681 = Rs 8405.

Q44. Correct Option: (a)
(a + b)(2a - 3b) = a(2a - 3b) + b(2a - 3b) = 2a^2 - 3ab + 2ab - 3b^2 = 2a^2 - ab - 3b^2.

Q45. Correct Option: (a)
Area = (1/2) x d1 x d2 => 240 = (1/2) x 16 x d2 => 240 = 8 x d2 => d2 = 30 cm.

Q46. Correct Option: (a)
Volume = 2m x 2m x 2m = 8 m^3. Since 1 m^3 = 1000 litres, capacity = 8 x 1000 = 8000 litres.

Q47. Correct Option: (a)
3^(-7 - (-10)) x 3^-5 = 3^3 x 3^-5 = 3^-2 = 1/9.

Q48. Correct Option: (a)
Inverse proportion: 12 x 40 = x x 30 => x = 480 / 30 = 16 workers.

Q49. Correct Option: (a)
49x^2 - 36 = (7x)^2 - (6)^2 = (7x - 6)(7x + 6).

Q50. Correct Option: (a)
Sum of digits = 2 + 4 + x = 6 + x. For divisibility by 3, 6+x must be multiple of 3. So x can be 0, 3, 6, 9.

SECTION B: INTELLIGENCE — SOLUTIONS

Q51. Correct Option: (a)
Each letter is moved one step forward (+1): P->Q, U->V, P->Q, I->J, L->M. TEACHER -> UFBDIFS.

Q52. Correct Option: (a)
Differences between consecutive terms are +4, +6, +8, +10. Next difference is +12 => 30 + 12 = 42.

Q53. Correct Option: (d)
Square, Rectangle, and Rhombus are 2D polygons, whereas Sphere is a 3D solid figure.

Q54. Correct Option: (a)
'My father's son' = himself (since he has no siblings). So 'that man's father' = himself. Thus, the photograph is of his son.

Q55. Correct Option: (a)
Replacing signs: 20 / 4 x 5 - 6 + 2 = 5 x 5 - 6 + 2 = 25 - 6 + 2 = 21.

Q56. Correct Option: (a)
A doctor works in a hospital; similarly, a teacher works in a school.

Q57. Correct Option: (a)
Walking N 5km, R (East) 3km, R (South) 5km brings him 3km East of the starting point.

Q58. Correct Option: (a)
Logical order from smallest to largest: Village (2) -> District (4) -> State (3) -> Country (1) -> Continent (5).

Q59. Correct Option: (a)
Rank from bottom = Total students - Rank from top + 1 = 40 - 15 + 1 = 26th.

Q60. Correct Option: (a)
Gaps between letters: +2 (A->C), +3 (C->F), +4 (F->J), +5 (J->O), +6 (O->U).

Q61. Correct Option: (a)
The word 'DICTIONARY' contains only one 'N', whereas 'NATION' requires two 'N's.

Q62. Correct Option: (a)
7^2 = 49; similarly 9^2 = 81.

Q63. Correct Option: (a)
Alphabetical values: D(4) + O(15) + G(7) = 26.

Q64. Correct Option: (a)
Dogs and Cats are distinct animals; both fall entirely inside the broader group of Animals.

Q65. Correct Option: (a)
First letter: Z, X, V, T (-2 steps -> R). Number: 1, 2, 6, 24 (x2, x3, x4, x5 -> 120). Last letter: A, D, G, J (+3 steps -> M). Next = R120M.

Q66. Correct Option: (a)
Cubes with 3 faces painted are located at the 8 corners of the large cube.

Q67. Correct Option: (d)
In all pairs except (14 - 198), the second number is the square of the first number (14^2 = 196, not 198).

Q68. Correct Option: (a)
R = 18, E = 5, D = 4 => 1854.

Q69. Correct Option: (a)
61 divided by 7 leaves a remainder of 5. Monday + 5 days = Saturday.

Q70. Correct Option: (a)
Angle at 3 o'clock = 3 x 30 degrees = 90 degrees.

Q71. Correct Option: (a)
Each step halves the circle; next shape is an arc or single point.

Q72. Correct Option: (a)
Mirror image reflects laterally from right to left.

Q73. Correct Option: (a)
A standard 5-pointed star contains 5 small triangles at the points + 5 large overlapping triangles = 10 triangles.

Q74. Correct Option: (a)
Pattern: a^2 + b^2 = c. 2^2 + 3^2 = 4 + 9 = 13. 4^2 + 5^2 = 16 + 25 = 41. 3^2 + 6^2 = 9 + 36 = 45.

Q75. Correct Option: (d)
Copper, Zinc, and Iron are pure element metals, whereas Brass is an alloy.

SECTION C: ENGLISH LANGUAGE — SOLUTIONS

Q76. Correct Option: (a)
The passage explicitly states that trees absorb carbon dioxide.

Q77. Correct Option: (a)
The passage states that forests prevent soil erosion.

Q78. Correct Option: (a)
Trees release oxygen.

Q79. Correct Option: (a)
'Importance of Trees' best reflects the theme of the passage.

Q80. Correct Option: (a)
'Accommodation' is the correct spelling with double 'c' and double 'm'.

Q81. Correct Option: (a)
Benevolent means well-meaning and kindly.

Q82. Correct Option: (a)
Pessimistic is the direct opposite of optimistic.

Q83. Correct Option: (a)
'Since' is used for a specific starting point in time in perfect continuous tense.

Q84. Correct Option: (a)
Past simple active ('repaired') changes to 'was/were + past participle' in passive ('was repaired').

Q85. Correct Option: (a)
Present continuous tense 'am reading' changes to past continuous 'was reading' in indirect speech.

Q86. Correct Option: (a)
'Must' expresses strong obligation or necessity.

Q87. Correct Option: (a)
'Where he lives' modifies the noun 'place', functioning as an Adjective Clause.

Q88. Correct Option: (a)
When subjects are joined by 'neither... nor', the verb agrees with the nearer subject ('his friends' -> plural 'were').

Q89. Correct Option: (a)
Positive sentence in simple present takes negative tag 'doesn't she?'.

Q90. Correct Option: (a)
'To spill the beans' means to disclose secret information prematurely.

Q91. Correct Option: (a)
'Useful' begins with a consonant sound /ju:/, so the indefinite article 'a' is used.

Q92. Correct Option: (a)
'Each' takes a singular verb 'has' and singular pronoun 'his'.

Q93. Correct Option: (a)
An optimist is a person who tends to be hopeful and confident about the future.

Q94. Correct Option: (a)
Opaque means not transparent or clear.

Q95. Correct Option: (a)
Universal truths and habitual actions are expressed in simple present tense ('rises').

Q96. Correct Option: (a)
Proper logical arrangement is 'Honesty is always the best policy'.

Q97. Correct Option: (a)
'Lest' is followed by 'should' to indicate fear or prevention of an undesirable outcome.

Q98. Correct Option: (a)
'Bravery' is an abstract noun representing a quality.

Q99. Correct Option: (a)
'Police' is a collective noun treated as plural in British/Indian English, taking 'are'.

Q100. Correct Option: (a)
A systematically organized list of books in a library is called a catalogue.

SECTION D: GENERAL SCIENCE — SOLUTIONS

Q101. Correct Option: (a)
Kharif crops (e.g., paddy, maize) are sown during the rainy season from June to September.

Q102. Correct Option: (a)
Lactobacillus bacteria promote the formation of curd from milk.

Q103. Correct Option: (a)
Rayon is obtained from wood pulp and resembles natural silk in appearance.

Q104. Correct Option: (a)
Mercury is the only metal that remains liquid at room temperature.

Q105. Correct Option: (a)
Metals react with acids to produce metal salts and hydrogen gas, which burns with a pop sound.

Q106. Correct Option: (a)
Petroleum and its derivatives are called 'black gold' due to their great commercial value.

Q107. Correct Option: (a)
Ignition temperature is the lowest temperature at which a substance catches fire.

Q108. Correct Option: (a)
Carbonisation is the slow process of conversion of dead vegetation into coal.

Q109. Correct Option: (a)
Biosphere reserves are large protected area units meant for conservation of biodiversity and cultural heritage.

Q110. Correct Option: (a)
Mitochondria generate ATP energy and are called the powerhouses of the cell.

Q111. Correct Option: (a)
Amoeba reproduces asexually by dividing into two equal daughter cells via binary fission.

Q112. Correct Option: (a)
Insulin secreted by the pancreas regulates glucose levels in the blood.

Q113. Correct Option: (a)
Electrostatic force is the non-contact attraction or repulsion between electric charges.

Q114. Correct Option: (a)
Lubricants like oil and grease reduce surface friction.

Q115. Correct Option: (a)
Loudness is proportional to the square of the amplitude of vibration.

Q116. Correct Option: (a)
Human hearing range spans frequencies between 20 Hz and 20,000 Hz.

Q117. Correct Option: (a)
Lemon juice contains citric acid which dissociates into ions and conducts electricity.

Q118. Correct Option: (a)
Electroplating deposits a thin protective or decorative metal coating using electrical current.

Q119. Correct Option: (a)
Fundamental rule of electrostatics: like charges repel, unlike charges attract.

Q120. Correct Option: (a)
The epicentre is the point on the surface directly above the underground focus of an earthquake.

Q121. Correct Option: (a)
Plane mirrors form virtual, erect, laterally inverted images of identical size to the object.

Q122. Correct Option: (a)
The Law of Reflection states that angle i = angle r.

Q123. Correct Option: (a)
Venus appears brightly in the sky before sunrise or after sunset and is called Morning/Evening Star.

Q124. Correct Option: (a)
The Moon is Earth's only natural satellite.

Q125. Correct Option: (a)
Carbon dioxide traps solar heat in the atmosphere, driving global warming.

SECTION E: SOCIAL SCIENCE — SOLUTIONS

Q126. Correct Option: (a)
Warren Hastings became the first Governor-General of Bengal in 1773.

Q127. Correct Option: (a)
The Battle of Plassey took place on 23 June 1757 between Siraj-ud-Daulah and Robert Clive.

Q128. Correct Option: (a)
Rajas and Taluqdars were recognized as zamindars responsible for collecting land revenue.

Q129. Correct Option: (a)
Nana Saheb, the adopted son of Peshwa Baji Rao II, led the revolt at Kanpur.

Q130. Correct Option: (a)
Raja Rammohan Roy founded the Brahmo Sabha (later Brahmo Samaj) in Calcutta.

Q131. Correct Option: (a)
The Rowlatt Act was passed in 1919, allowing detention of political prisoners without trial.

Q132. Correct Option: (a)
The Civil Disobedience Movement began with the breaking of the Salt Law at Dandi in 1930.

Q133. Correct Option: (a)
Ubiquitous resources are present everywhere, whereas localized resources occur in specific places.

Q134. Correct Option: (a)
Topsoil (A-horizon) contains high organic matter and humus.

Q135. Correct Option: (a)
Terrace farming cuts flat steps on steep slopes to reduce surface runoff and soil erosion.

Q136. Correct Option: (a)
Mixed farming combines crop cultivation with animal husbandry.

Q137. Correct Option: (a)
The Chota Nagpur Plateau is rich in iron ore and coal reserves, housing major steel industries.

Q138. Correct Option: (a)
The Supreme Court of India was established on 26 January 1950.

Q139. Correct Option: (a)
Article 17 explicitly abolishes untouchability and forbids its practice in any form.

Q140. Correct Option: (a)
Following constitutional amendments, the maximum strength of Lok Sabha is 550 members.

Q141. Correct Option: (a)
The Vice-President of India serves as the ex-officio Chairman of Rajya Sabha.

Q142. Correct Option: (a)
Article 21 states that no person shall be deprived of life or personal liberty except by procedure established by law.

Q143. Correct Option: (a)
PIL allows any public-spirited individual to approach the court on behalf of marginalized communities.

Q144. Correct Option: (a)
Shifting cultivation is called Jhumming in Assam, Meghalaya, and Nagaland.

Q145. Correct Option: (a)
Iron and steel provides raw material and machinery for almost all other manufacturing sectors.

Q146. Correct Option: (a)
Jyotirao Phule dedicated 'Gulamgiri' to the American anti-slavery movement.

Q147. Correct Option: (a)
The 86th Constitutional Amendment Act (2002) inserted Article 21A granting right to education for ages 6-14.

Q148. Correct Option: (a)
Both regions are world-leading hubs for Information Technology (IT) and software development.

Q149. Correct Option: (a)
Lord Dalhousie annexed states like Satara, Sambalpur, and Jhansi using the Doctrine of Lapse.

Q150. Correct Option: (a)
Rajya Sabha members are elected for a term of 6 years, with one-third retiring every 2 years.